Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/01/2004
PI Name:	Graves, Lee M Ph.D.		
Project Title:	Application Of Proteomics To Study Skeletal Muscle Atrophy Resulting From Exposure To Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	LMG@med.unc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919/966-0915
Organization Name:	University of North Carolina at Chapel Hill		
PI Address 1:	Department of Pharmacology		
PI Address 2:	936 MEJ Bldg, PO Box 7365		
PI Web Page:			
City:	Chapel Hill	**State:**	NC
Zip Code:	27599-7365	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/15/2002	**End Date:**	11/14/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1599		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose to use modern techniques of mass spectrometry and "proteomics" to analyze both global and specific changes in protein expression during muscle atrophy. Analogous to "gene chips," this approach will allow us to identify many of the key enzymes affected by muscle wasting. In addition, we will focus on characterizing the change in tyrosine phosphorylation of key intracellular signaling proteins and identifying the target signaling molecules influenced during this process. Finally the ability of growth stimulatory factors or hormones to reverse this process of muscle atrophy will be examined. These studies will thus be instrumental in identifying the changes that occur in skeletal muscle upon wasting and will provide fundamental information required for the design of therapeutic agents useful against this condition. To determine how skeletal muscle atrophy influences the development of insulin resistance through changes in the regulation of protein kinase signaling. Important metabolic change during atrophy that may influence the function of skeletal muscle in astronauts exposed to low gravity conditions.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Not Available
Task Progress:	We have made substantial progress investigating the biochemical basis for the metabolic changes in skeletal muscle during the development of atrophy. Our model system has been the rat hindlimb suspension system carried out in collaboration with Drs. Wade and Grindeland at the NASA Ames Research Center. Two major studies have been completed that involved both a time course analysis and a comparison of the effects of growth hormone on the development of atrophy. What we discovered was a significant activation of an important cell regulatory kinase c-Jun kinase, in response to hindlimb suspension. Because of the fundamental role of this kinase in regulating an insulin-regulated response (through phosphorylation of IRS-1), we investigated the effects of atrophy on IRS-1 phosphorylation. Interestingly our studies demonstrated that in response to atrophy-dependent increases in JNK, IRS-1 was phosphorylated on a specifc serine residue (307). This phosphorylation is known to target IRS-1 for degradation and our results confirmed that IRS-1 was rapidly degraded as a result of the induction of atrophy. (This research was published Hilder et al., 2003). We have recently extended these studies to include a time course analysis of atrophy from 2-13 weeks and confirmed our original observations showing the increase in JNK activity and the degradation of IRS-1. In this study we compared the effects of hypergravity; however, we find that although hypergravity induces muscle loss similar to atrophy, no change in JNK or IRS-1 was observed. Thus the changes in IRS-1 appear to be atrophy specific. We also compare the regulation of the key glucose transporters (GLUT1/4) during atrophy. We find a substantial increase in the expression of these proteins. To further examine the kinases necessary for these changes we examine the phosphorylation and regulation of a number of others implicated in GLUT1/4 regulation. In these studies we observed significant atrophy-dependent changes in activity of AMP kinase, Erk, P38 MAPK and PYK-2. These studies are the basis of a manuscript that we just completed (Hilder et al., 2004). We are also continuing to develop and apply proteomics to study changes in protein expression and phosphorylation during muscle atrophy. Our overall goal is to obtain a comprehensive analysis of the proteins and phosphorylation events that define changes in muscle metabolism after exposure to atrophy.
Bibliography Type:	Description: (Last Updated: 10/21/2009)
Articles in Peer-reviewed Journals	Hilder TL, Baer LA, Fuller PM, Fuller CA, Grindeland RE, Wade CE, Graves LM. "AMP-activated protein kinase, Erk 1/2 and p38 MAP kinase may cooperate to enhance glucose uptake in insulin-resistant atrophic skeletal muscle." Manuscript submitted, January 2005. , Jan-2005
Articles in Peer-reviewed Journals	Hilder TL, Tou JC, Grindeland RE, Wade CE, Graves LM. "Phosphorylation of insulin receptor substrate-1 serine 307 correlates with JNK activity in atrophic skeletal muscle." FEBS Lett. 2003 Oct 9;553(1-2):63-7. PubMed PMID: 14550547 , Jan-2004